Physiology and Geometry-Informed Deep Learning Framework for 12-Lead ECG Cardiac Transmembrane Potential recovery

Shujin Hu
State Key Laboratory of Extreme Photonics and Instrumentation, College of Optical Science and Engineering, Zhejiang University, Hangzhou, 310027, China

Huihui Ye
State Key Laboratory of Extreme Photonics and Instrumentation, College of Optical Science and Engineering, Zhejiang University, Hangzhou, 310027, China

Huafeng Liu*
State Key Laboratory of Extreme Photonics and Instrumentation, College of Optical Science and Engineering, Zhejiang University, Hangzhou, 310027, China

ABSTRACT

The recovery of cardiac transmembrane potential (TMP) from body surface potential (BSP) or so-called electrocardiogram imaging (ECGI) is a critical non-invasive reconstruction technique with significant implications for clinical guidance in cardiac disease diagnosis. Existing non-invasive methods often rely on electrocardiogram (ECG) from large number of leads (64-256 leads), of which is not as affordable and user-friendly as 12-lead ECG. In this study, we propose to use 12-lead ECG to recover patient-specific TMP patterns with our proposed physiology and geometry-informed deep learning framework for accurate non-invasive reconstruction, offering a more cost-effective and convenient alternative as compared to current methods. Specifically, we introduce a deep learning framework that integrates attention mechanisms and graph convolution network (GCN) to learn patient-specific TMP temporal and spatial characteristics. Additionally, we introduce a transfer matrix H, which is derived from the specific geometric model, to facilitate the inverse reconstruction of TMP. Through a series of simulation experiments, our method shows low mean square error (MSE) of 0.018, high correlation coefficient (CC) of 0.952, and substantial dice correlation (DC) of 0.676, which validate our model in accurately reconstructing patient-specific TMP from 12-lead ECG data. We also validate our method on 6 patient (with myocardial infarction) data and it shows good accordance in infraction detection from our proposed method and that with MRI.

CCS CONCEPTS

• **Applied computing**; • **Life and medical sciences**; • **Health care information systems**;

KEYWORDS

Cardiac electrophysiological reconstruction, 12-lead ECG, Dynamic physiological model, Inverse problem

*Huafeng Liu is the corresponding author. Author contact: liuhf@zju.edu.cn

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICIAI 2024, March 16–18, 2024, Tokyo, Japan
© 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-0930-2/24/03
https://doi.org/10.1145/3655497.3655505

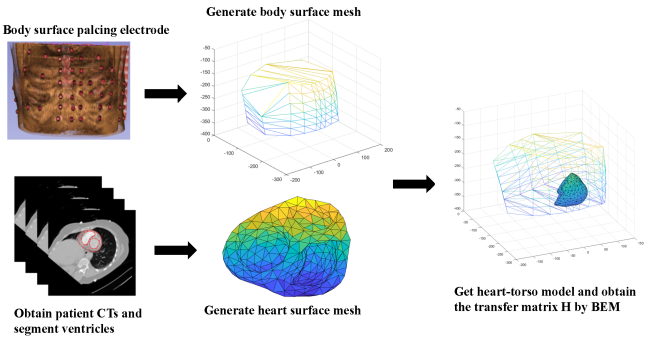


Figure 1: A pipeline showing the generation of patient-specific heart-torso model and acquisition of transfer matrix H.

ACM Reference Format:
Shujin Hu, Huihui Ye, and Huafeng Liu. 2024. Physiology and Geometry-Informed Deep Learning Framework for 12-Lead ECG Cardiac Transmembrane Potential recovery. In *2024 the 8th International Conference on Innovation in Artificial Intelligence (ICIAI 2024), March 16–18, 2024, Tokyo, Japan.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3655497.3655505

1 INTRODUCTION

The non-invasive retrieval of TMP from BSP provides a detailed and comprehensive insight into cardiac electrophysiological activity, playing a crucial role in clinical settings. This information aids in diagnosing arrhythmias and facilitates localizing ischemic or scar tissue [1]. Current techniques, such as Electrocardiographic Imaging (ECGI) [2] typically use a series of cardiac medical image (such as CT) and electrode patch positions (ranging from 64 to 256 [3]) to construct a personalized heart-torso model [4, 5]. The mapping relationship from the heart to the body surface is obtained using the boundary element method, represented by the transfer matrix H, Figure 1 describes the specific heart-torso modeling process. Electrophysiology can then be simulated based on this model for further research.

Retrieving TMP from BSP as an inherent ill-posed problem is highly challenging [6], necessitating additional constraints to ensure a unique and physiologically meaningful solution. Various approaches have been proposed to address this issue. One method integrates mathematical iteration and deep learning methods to introduce these constraints [7-9], while another employs statistical inference to resolve inherent uncertainties [10-11]. Current

research methodologies impose physiological constraints on the inverse problem by considering cardiac electrodynamics, such as [12-14]. However, these approaches often utilize the transfer matrix as a constraint within the loss function rather than as a priori guidance for the inverse reconstruction process combined with neural network.

Another challenge lies in gathering and processing a large number of signals, which augment workload for doctors and researchers. To mitigate this challenge, we opt for the 12-lead ECG data for reconstruction, which is a more widely utilized and cost-effective alternative in clinical practice. However, it has lower spatial resolution and more limited information compared to multi-lead devices.

To address these challenges, we propose a physiology and geometry-informed deep learning framework that integrates the transfer matrix H with an Inverse-block as a guide for the inverse reconstruction of TMP. Additionally, we employ a physiological model prior to simulate an extensive dataset, providing guidance for the network framework during the learning process. Furthermore, we incorporate spatial and temporal feature encoding blocks that leverage the geometric structure of the heart to learn specific TMP patterns in 3D dynamics. To tackle ill-posedness, Laplacian smoothing regularization is introduced.

Our contributions can be summarized in three key aspects:

- We introduce a TMP recovery method based on 12-lead ECG, offering a more affordable and user-friendly alternative to guide clinical diagnosis of cardiovascular diseases.
- We present a patient-specific deep learning network consisting of three key components: i) an inverse reconstruction block utilizing the transfer matrix H as a prior, ii) a spatial feature encoding block that combines graph convolutional networks (GCN) with nonlocal spatial attention, and iii) a temporal attention block that aid with the spatial block for surrogate physiological modeling.
- Our network is evaluated using both synthetic and real data. Through experiments with synthetic data, we illustrate our network's capability to reconstruct dynamic TMP distributions. Moreover, in the task of myocardial infarct detection, we achieved high detection accuracy.

2 METHODOLOGY

2.1 Problem definition

The forward relationship between TMP and BSP can be modeled as the following linear equation by applying the quasi-static approximation of electromagnetic theory [15]:

$$Y = H\mathrm{u}$$

in which $Y \in \mathbb{R}^{\mathrm{n}\times t}$,$u \in \mathbb{R}^{m\times t}$are the BSP and TMP respectively. $H \in \mathbb{R}^{\mathrm{n}\times m}$is the transfer matrix which contains the geometry and conductivity information of the heart-torso model. m corresponds to the number of heart nodes, and $n$denotes the number of leads.

The traditional TMP reconstruction problems try to seek an inverse solution u that both minimizing the data consistency with the observational data Y while satisfying certain constraints$\Re(u)$ [16]:

$$\hat{u} = \arg\min \|Y - Hu\|_2^2 + \lambda\Re(u)$$

The regularization parameter λ is often chosen empirically. In this work, we introduce the Laplacian smoothing over the heart geometry as the second term.

2.2 Physiological model introduced in our framework

Cardiac electrophysiology (EP) models account for the spatial distribution of cardiac potentials and their propagation, offering a comprehensive representation of the heart's electrical activity. In our study, we focus on using a simple bivariate potential-based model known as the FitzHugh-Nagumo (FHN) model [17]. The process of cardiac excitation can be described as follows:

$$\frac{du}{dt} = \nabla \cdot (D\nabla \mathrm{u}) + f_1(u, v)$$

$$\frac{dv}{dt} = f_2(u, v)$$

where u stands for TMP and v for recovery current. D is diffusion tensor. Variations of $f_1(u, v)$and $f_2(u, v)$produce different TMP shapes. The associated boundary condition$\frac{du}{d\vec{n}}$assumes no active current leaving heart surfaces ($\vec{n}$ is the outward normal vector of boundary surfaces).

We generate a large number of data pairs through ECGSim simulation software [18] by modifying the corresponding parameters based on the EP model. This dataset serves as the training data for our framework aimed at capturing the dynamic characteristics of cardiac electric activity.

2.3 Description of the different blocks of the network

2.3.1 Inverse-block. We consider the inverse process as a variant of the forward process. Therefore, we propose to reconstruct TMP using the H matrix as a prior, which can be calculated as follows:

$$X^0 = \sum_{i=1}^{d} soft\max(\frac{Q_i^H (K_i^Y)^T}{\sqrt{d}}) V_i^Y$$

We start by computing the pseudo-inverse of $H \in \mathbb{R}^{\mathrm{n}\times m}$and subsequently perform linear transformations at each time step, yielding the vector$Q^H \in \mathbb{R}^{\mathrm{t}\times m\times d}$. Here, $d$represents the dimension. The vectors $K^Y \in \mathbb{R}^{\mathrm{t}\times n\times d}$ and $V^Y \in \mathbb{R}^{\mathrm{t}\times n\times d}$ are derived from ECG data $Y^0 \in \mathbb{R}^{n\times t}$ through a series of linear transformation layers. Figure 2 shows the overall structure of proposed network.

At each time step, we calculate the inner product of the$Q^H$and K^0vectors to model the feature relationship between BSP and$H$, generating an attention matrix that aids in reconstructing the TMP vector$X^0 \in \mathbb{R}^{\mathrm{t}\times m\times d}$. To normalize attention scores within the matrix, we apply the softmax function. As depicted in Figure 2, our cross-attention module corresponds to the Inverse-Block.

The mentioned process is concurrently carried out across multiple subspaces, enhancing the model's capacity to learn diverse features.

2.3.2 Spatial encoder block. We have developed a geometry-informed spatiotemporal feature encoding block by integrating

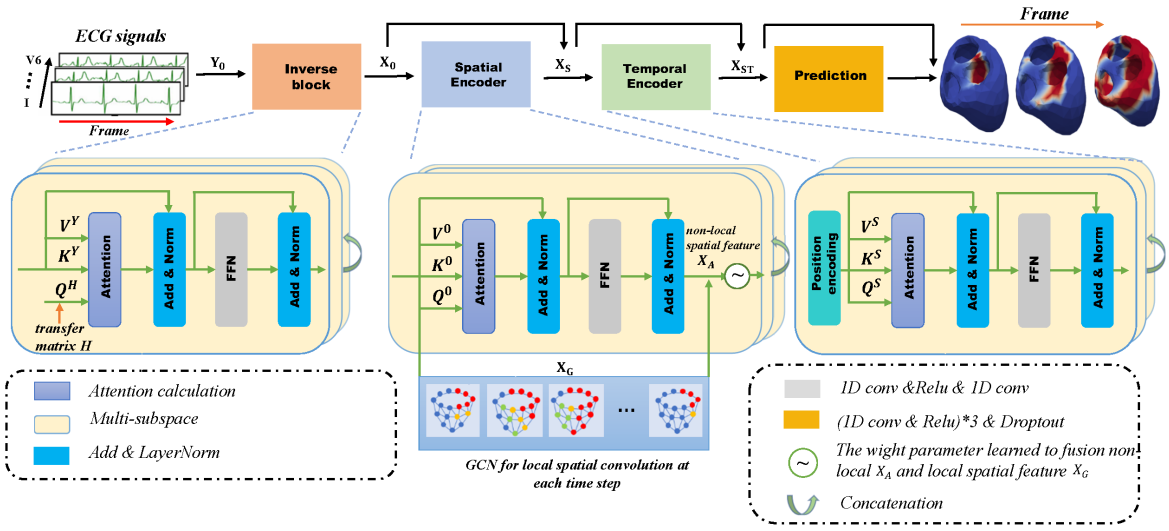


Figure 2: Illustrative architecture of the proposed physiology and geometry-Informed deep learning network. It comprises an Inverse-block, stacked spatial-temporal encoding blocks, and a prediction layer. The Inverse-block incorporates the transfer matrix H to guide the reconstruction process. The spatial encoding block combines a spatial transformer with GCN to collectively capture spatial dependencies. Meanwhile, the temporal encoding block focuses on temporal dependencies

GCN with attention mechanism. We represent the heart's surface as a triangular mesh, which is then translated into an undirected graph. To establish connections, we normalize the properties of edges based on their Euclidean distances. Given initial node features $X^0 \in \mathbb{R}^{t\times m\times d}$, the graph convolution operation aggregates information for each node and its neighbors at each timestep by the following equation:

$$X_G = \sigma(LX^0W) \tag{1}$$

Here, $\sigma(\cdot)$ is the activation function, L denotes the normalized Laplacian matrix, and W stands for the network layer's parameters. The use of GCN enables us to capture local spatial features effectively by considering relationships between nodes and their neighbors.

Given the complex structure of the heart, integrating global information through multi-layer GCN stacking is crucial. However, this approach requires significant computational resources. To address this challenge, we propose a spatial attention module that automatically assigns weights to all nodes within the heart and extracts non-local spatial information. The spatial attention is calculated using the following equation:

$$X_A = \sum_{i=1}^{d} soft\max(\frac{Q_i^0(K_i^0)^T}{\sqrt{d}})V_i^0 \tag{2}$$

Vectors Q^0,K^0 and V^0 represent the results of different linear transformations of the output from the Inverse-Block. To combine local and non-local information, we have devised a weighting mechanism, which can be calculated as follows:

$$g = sigmoid(f_g(X_G) + f_a(X_A)) \tag{3}$$

$$X_S = g(X_G) + (1-g)X_A \tag{4}$$

where f_g and f_a are linear projections that transform X_G and X_A into 1-D vectors respectively. As a result, the output X_S is obtained by weighting X_G and X_A with the weight parameter.

2.3.3 Temporal encoder block. The temporal attention module is cascaded with the spatial attention module, producing the feature encoding vector X_{ST} which captures spatiotemporal dependencies:

$$X_{ST} = \sum_{i=1}^{d} soft\max(\frac{Q_i^S(K_i^S)^T}{\sqrt{d}})V_i^S \tag{5}$$

Here, Q^S, K^S, V^S are derived from the X_S after linear transformation. To simultaneously integrate temporal relative information during parallel computing, we adding position encoding [19] with the input vectors.

3 EXPERIMENT

3.1 Experiment setting

3.1.1 Data acquisition. In our study, we employed the ECGSim software to generate simulated cases of MI and ectopic pacing for training network. The heart's biventricular surface consisted of 697 nodes, and the transfer matrix H between the heart and torso model was established using a boundary element model. Various scenarios were simulated by adjusting depolarization and repolarization timing, as well as amplitude and slope parameters of TMP.

Myocardial infarction cases involved reducing the amplitude of TMP in specific areas with infarct radii randomly chosen within a range of 2mm to 10mm, establishing both transmural and non-transmural infarcts separately. Additionally, cases of ectopic pacing on different nodes were included. Approximately 10,000 sets of TMP-BSP data (6,745 for infarction and others for ectopic pacing) were simulated with BSP augmented by 30dB Gaussian noise. The

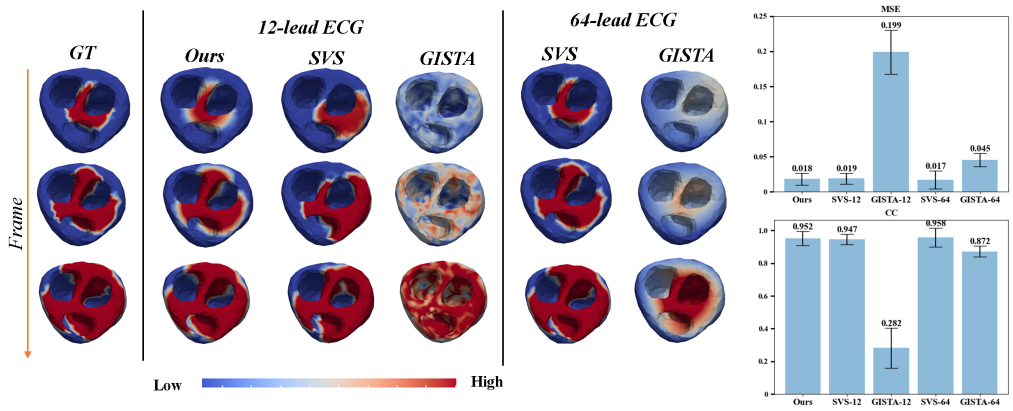


Figure 3: (Left) Comparison of visualization results of dynamic TMP recovery by different methods. Red is the actived area, blue is the resting area. (Right) Quantitative analysis results of TMP reconstructed by different methods.

dataset was divided into training, validation, and test sets in an 8:1:1 ratio.

The clinical data comprises 6 cases of myocardial infarction, with BSP data obtained through 12-lead ECG sampling at a frequency of 2000Hz. The corresponding labels were acquired via delayed enhancement MRI.

Preprocessing of the clinical ECG data involves removing BSP noise using a symlet wavelet filter and correcting baseline drift by fitting a low-order polynomial to the signal and subtracting it. The data range is normalized to be consistent with the training data.

3.1.2 Measurement metrics. Three metrics are used to evaluate the performance of the reconstruction: 1) mean square error (MSE) , 2) correlation coefficient (CC), and 3) dice coefficient$DC = \frac{2(|S_1 \cap S_2|)}{|S_1 + S_2|}$, where $S_1$and $S_2$are the sets of heart surface nodes in the ground truth and estimated regions of the infarcted area, respectively.

3.1.3 Loss function definition. The loss function in our work is defined as follows:

$$\widehat{u} = \arg\min \|Y - Hu\|_2^2 + \lambda \|Lu\|_2^2$$

L is the normalized Laplace matrix, $\lambda$is empirically tuned.

3.1.4 Other details. To validate the feasibility of our proposed TMP recovery method, we conducted the following experiments:

- We tested our method on two tasks: restoring the spatiotemporal propagation characteristics of TMP and detecting myocardial infarction (MI) areas. Furthermore, we compared our method with two other deep learning-assisted TMP recovery methods, namely SVS [20] and GISTA [7]. These methods were applied to both 12-lead and 64-lead ECG data for comparative analysis.
- To validate the network blocks, we extracted and visualized the attention matrix of each block.
- Finally, we utilized six clinical cases of MI to validate the clinical applicability of our approach.

3.2 Phantom experiments

3.2.1 Dynamic TMP recovery. Figure 3 illustrates the visualization (left) and quantification (right) results of various methods for recovering dynamic TMP. The red area is the activated area, and the blue area is the resting area. Our proposed 12-lead-based approach achieves closer reconstruction to the ground truth (GT). While the SVS method misidentifies activation locations with 12-lead data, its performance improves with 64-lead data. Our method can attain comparable results to the 64-lead approach at a slight sacrifice in accuracy. Conversely, the GISTA-based method struggles to fully reconstruct dynamic TMP, particularly pronounced with 12-lead data.

3.2.2 Myocardial infarction detection. The results of myocardial infarction detection using various methods are depicted in Figure 4, showing different types of infarctions such as endocardial, epicardial, and transmural cases. The red area in the figure represents the infarct region. Our proposed method closely reproduces the ground truth when utilizing 12-lead measurement data.

In Figure 5, the left part displays a biventricular bull's eye diagram with the left ventricle divided into 10 parts and the right ventricle into 5 parts as described in [21]. Our method shows high accuracy in detecting apical-anterior, apical-septal, and apical-lateral regions (7, 8, 10). For right ventricular infarction, our method can only detect the region near the heart's apex. The right-hand side of Figure 5 presents statistical results for infarct detection using different methods, demonstrating that our approach offers advantages when using less measurement data.

3.2.3 Effect verification of different blocks. To evaluate the impact of each block, we visualized the attention matrix (Figure 6), where brighter cells indicate higher attention. Within this visualization, we observed notably higher attention on leads III and V5 in the Inverse-Block(a). (b) is the adjacency matrix. Spatial attention (c-e) captures sparse spatial features. We selected both infarct and healthy node for visualization their TMP (i), represented by (f). Their temporal attention matrices are shown in (g) and (h) respectively. In healthy node, attention concentrates on both depolarization and early repolarization phases; however, for infarct

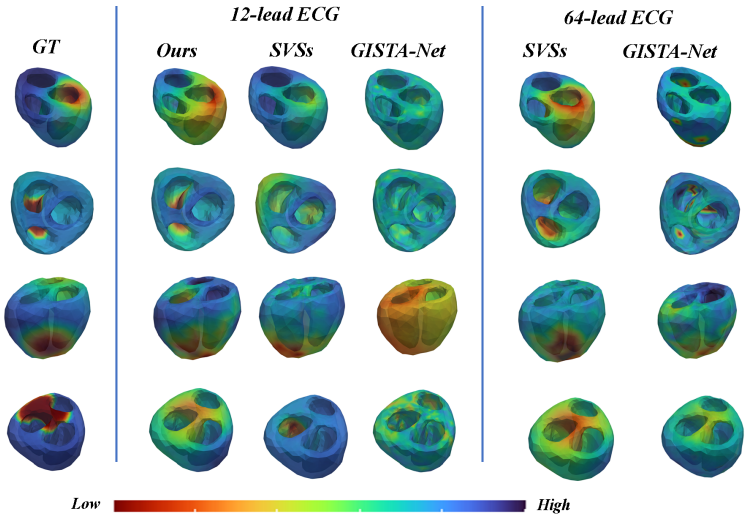


Figure 4: Four cases of myocardial infarction (MI) occurring at distinct sites, encompassing endocardial, epicardial, and transmural infarctions, were utilized to evaluate various methods at different measurements. The infarct area is visually represented by the red region.

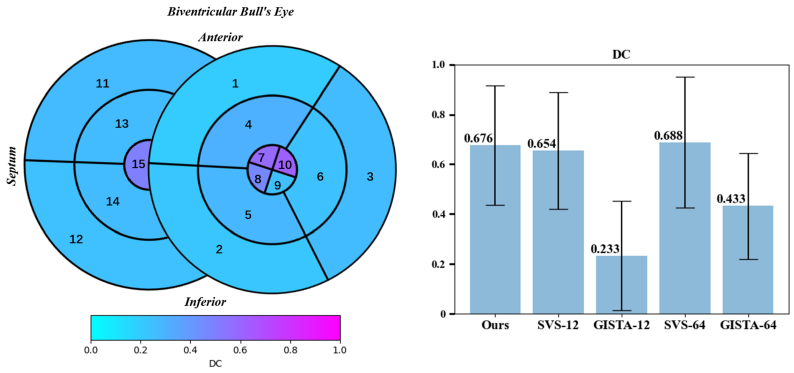


Figure 5: (Left) DC test results of MI area in different parts of the biventricle. (Right) Statistical quantification of infarction detection by different method.

Table 1: Quantitative analysis results of ablation experiment.

	MSE	CC	DC
Baseline	0.069	0.806	0.312
Baseline+SE	0.075	0.751	0.286
Baseline+TE	0.063	0.812	0.353
Baseline+TE+SE	**0.018**	**0.952**	**0.676**

node, the model focuses on early repolarization phases with less attention weight. Moreover, we utilized Inverse-block as a baseline to evaluate model enhancement through separate spatial and temporal modules. The result shows in Table 1. Our findings indicate that incorporating a temporal block improves baseline performance; however, optimal model performance is achieved only when both temporal and spatial blocks introduced.

3.3 Clinical data experiment

We tested our framework's clinical applicability on 6 myocardial infarction(MI) cases. Figure 7 shows detection results, with blue areas marking detected infarcts regions which correlates well with that from MRI. Our findings demonstrate that our network allowing rough localization of infarct areas; this verification underscores the network's potential for preliminary clinical MI detection.

4 CONCLUSION AND DISCUSSION

We proposed a deep learning-based framework to non-invasively reconstruct TMP from 12-lead ECG. Patient-specific spatiotemporal features of TMP are learned by combining the attention mechanism and GCN, and a specific geometric model transfer matrix is introduced to facilitate the inverse reconstruction of TMP. Through a series of simulation experiments, the effectiveness of the method

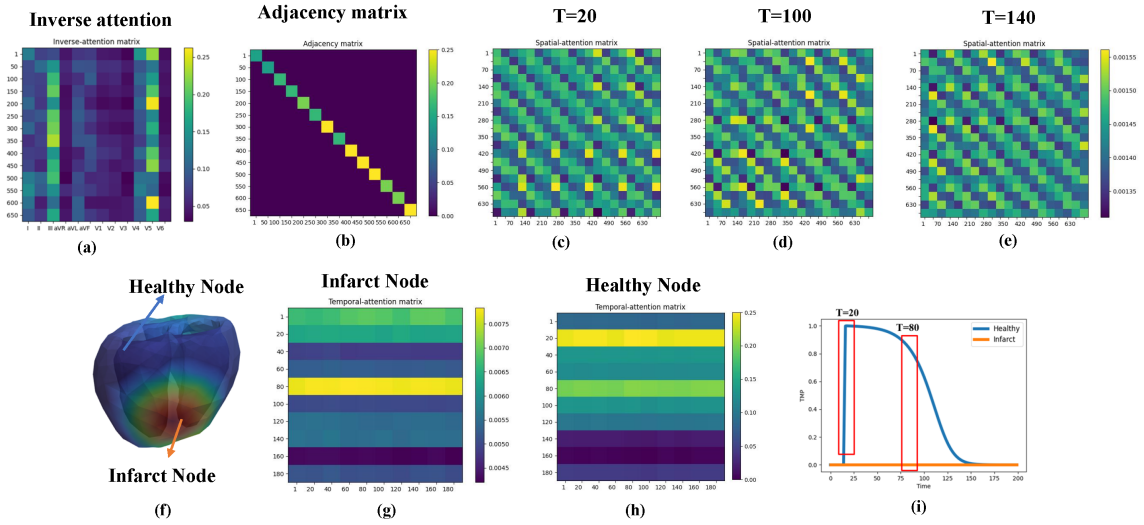


Figure 6: Visualization of attention matrix:(a) Inverse-block's attention, (b) the adjacency matrix of the heart model. (c-e) spatial attention at timestep 20ms,100ms,140ms, (f) optional health node and infarction node chosen for temporal attention visualization, (g-h) is their temporal attention respectively and corresponding moments on TMP (i).

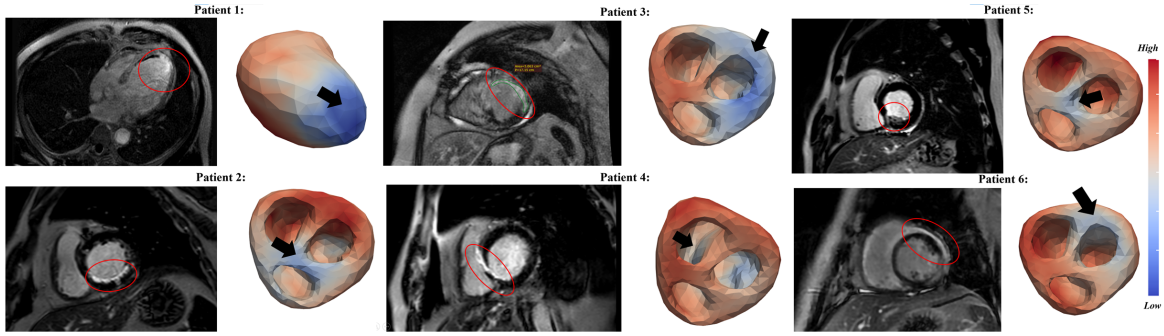


Figure 7: Six clinical cases of MI, with MRI results on the left and reconstructed infarction areas in blue on the right.

was demonstrated and verified on data of patients suffering from myocardial infarction, showing good consistency with MRI.

ACKNOWLEDGMENTS

This work is supported in part by the Talent Program of Zhejiang Province (No: 2021R51004), the National Natural Science Foundation of China (No: U1809204, 61525106).

REFERENCES

[1] U Rajendra Acharya, Hamido Fujita, Shu Lih Oh, Yuki Hagiwara, Jen Hong Tan, and Muhammad Adam. 2017. Application of deep convolutional neural network for automated detection of myocardial infarction using ECG signals. Information Sciences 415 (2017), 190–198.

[2] Raja N Ghanem, Ping Jia, Charulatha Ramanathan, Kyungmoo Ryu, Alan Markowitz, and Yoram Rudy. 2005. Noninvasive electrocardiographic imaging (ECGI): comparison to intraoperative mapping in patients. Heart Rhythm 2, 4 (2005), 339–354.

[3] Sandesh Ghimire, John L. Sapp, B. Milan Horáček, and Linwei Wang. 2019. Noninvasive Reconstruction of Transmural Transmembrane Potential With Simultaneous Estimation of Prior Model Error. IEEE Transactions on Medical Imaging 38, 11 (2019), 2582–2595.

[4] Helder Pereira, Steven Niederer, and Christopher A Rinaldi. 2020. Electrocardiographic imaging for cardiac arrhythmias and resynchronization therapy. EP Europace 22, 10 (08 2020), 1447–1462.

[5] Linwei Wang, Heye Zhang, Ken C. L. Wong, Huafeng Liu, and Pengcheng Shi. 2010. Physiological-Model-Constrained Noninvasive Reconstruction of Volumetric Myocardial Transmembrane Potentials. IEEE Transactions on Biomedical Engineering 57, 2 (2010), 296–315.

[6] R.M. Gulrajani. 1998. The forward and inverse problems of electrocardiography. IEEE Engineering in Medicine and Biology Magazine 17, 5 (1998), 84–101.

[7] Lide Mu and Huafeng Liu. 2021. Cardiac transmembrane potential imaging with GCN based iterative soft threshold network. In Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part VI 24. Springer, 547–556.

[8] Linsheng Cheng and Huafeng Liu. 2021. Noninvasive cardiac transmembrane potential imaging via global features based FISTA network. In 2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC). IEEE, 3149–3152.

[9] Shuting Xie, Linsheng Cheng, and Huafeng Liu. 2021. Lnista-Net: learning nonlocal regularized iterative shrinkage-thresholding network for noninvasive cardiac transmembrane potential imaging. In 2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI). IEEE, 183–186.

[10] Sandesh Ghimire, Prashnna Kumar Gyawali, Jwala Dhamala, John L Sapp, Milan Horacek, and Linwei Wang. 2019. Improving generalization of deep networks for inverse reconstruction of image sequences. In Information Processing in Medical Imaging: 26th International Conference, IPMI 2019, Hong Kong, China, June 2–7, 2019, Proceedings 26. Springer, 153–166.

[11] Sandesh Ghimire, John L. Sapp, B. Milan Horáček, and Linwei Wang. 2019. Noninvasive Reconstruction of Transmural Transmembrane Potential With Simultaneous Estimation of Prior Model Error. IEEE Transactions on Medical Imaging 38, 11 (2019), 2582–2595.

[12] Yubo Ye, Huafeng Liu, Xiajun Jiang, Maryam Toloubidokhti, and Linwei Wang. 2023. A Spatial-Temporally Adaptive PINN Framework for 3D Biventricular Electrophysiological Simulations and Parameter Inference. In International Conference on Medical Image Computing and Computer-Assisted Intervention. Springer, 163–172.

[13] Bing Yao and Hui Yang. 2020. Spatiotemporal regularization for inverse ECG modeling. IISE Transactions on Healthcare Systems Engineering 11, 1 (2020), 11–23.

[14] Bing Yao and Hui Yang. 2016. Physics-Driven Spatiotemporal Regularization for High-dimensional Predictive Modeling: A Novel Approach to Solvethe Inverse ECG Problem. Scientific Reports 6 (Dec. 2016), 39012.

[15] Kirsten HWJ Ten Tusscher and Alexander V Panfilov. 2006. Alternans and spiral breakup in a human ventricular tissue model. American Journal of Physiology-Heart and Circulatory Physiology 291, 3 (2006), H1088–H1100.

[16] Andrei Nikolaevich Tikhonov. 1963. On the solution of ill-posed problems and the method of regularization. In Doklady akademii nauk, Vol. 151.Russian Academy of Sciences, 501–50.

[17] Christopher KRT Jones. 1984. Stability of the travelling wave solution of the FitzHugh-Nagumo system. Trans. Amer. Math. Soc. 286, 2 (1984), 431–469.

[18] A Oosterom and Thom Oostendorp. 2004. ECGSIM: An interactive tool for studying the genesis of QRST waveforms. Heart (British Cardiac Society) 90 (03 2004), 165–8.

[19] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention isall you need. Advances in neural information processing systems 30 (2017).

[20] Xiajun Jiang, Maryam Toloubidokhti, Jake Bergquist, Brian Zenger, Wilson W Good, Rob S MacLeod, and Linwei Wang. 2022. Improving Generalization by Learning Geometry-Dependent and Physics-Based Reconstruction of Image Sequences. IEEE Transactions on Medical Imaging 42, 2 (2022), 403–415.

[21] Eduardo Jorge Godoy, Miguel Lozano, Ignacio M García-Fernández, Ana Ferrer-Albero, Rob Macleod, Javier Saiz, and Rafael Sebastian. 2018. Atrial Fibrosis Hampers Non-invasive Localization of Atrial Ectopic Foci From Multi-Electrode Signals: A 3D Simulation Study. Frontiers in Psychiatry 9/404 (2018).